3.11. Finding and correcting errors

As formulas get more complex, the chances of errors creeping in become higher. Some errors will announce themselves to you with some error condition; others may not be so simple—you may have given an incorrect reference that will make for a wrong result.

When Excel itself reports errors with the formula, they are relatively easy to correct. There are different types of standard error conditions that will give you a good indication of where exactly to look to correct the error.

First, however, you have to take certain precautions when entering formulas:

- Ensure that the brackets are properly matched. This is especially a problem with nested formulas (one formula inside another which again maybe inside another formula). Take care to ensure that all the brackets are correctly used. Excel will normally catch any bracket errors, but it can miss them when the formula is complex. The maximum level of nesting in any case is 64.

- Cell ranges are indicated by colons. The cell range A1 to A23 should be entered as A1:A23, not A1-A23. If you use A1-A23, Excel will subtract the value of A23 from A1.

- Ensure that all the required values (or “arguments” as they are called) are entered into the formula. Each formula may have required values and optional values; at the minimum, the required values have to be entered to prevent Excel from bringing up an error.

- If you refer to cells in other worksheets, ensure that the worksheet name is enclosed in single quotes. If you are referring to links in an external workbook, ensure that you have entered the full path to that workbook.

Excel error checking is rule-based, and it tries to learn your preferences as it goes. Any errors which you ignore, Excel will also